Portable ECG Holter Prototype

Final-year project — Professional Bachelor's Degree
Biomedical Equipment Maintenance
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Revised version for project archiving and presentation

Abstract

This project focuses on the design of a **portable ECG Holter prototype** for signal acquisition, conditioning and wireless transmission.

The main work concerned the analog acquisition chain, amplification using an **AD620**, reduction of **50 Hz power-line interference** using a Notch filter, and Bluetooth transmission to a smartphone. The mobile application developed with MIT App Inventor was designed not only for visualization but also, as a future extension, for storing ECG curves throughout the Holter monitoring period.

The prototype was experimentally evaluated using controlled electrical signals supplied by a generator through three inputs corresponding to the three electrode inputs of the circuit. The acquisition chain and smartphone visualization were demonstrated. Human ECG acquisition was not performed.

1. Introduction and Project Context

A conventional ECG provides an observation of cardiac electrical activity over a limited period. This can be insufficient when symptoms are intermittent and do not occur during the examination. The Holter principle addresses this limitation by enabling ambulatory monitoring over a longer period.

The purpose of this project was therefore to develop an electronic prototype combining an analog conditioning stage, digital acquisition and wireless communication. The smartphone was considered part of the system from the beginning: first as a visualization interface and, in the intended evolution of the project, as a storage medium for ECG curves throughout the monitoring period.

The practical work did not aim to build a clinically validated automatic ECG interpretation system. The main focus was the electronic acquisition chain, the study of 50 Hz filtering, Bluetooth communication and mobile visualization.

2. Project Objectives

Main objective: develop an ECG Holter prototype able to acquire a conditioned electrical signal, transmit it wirelessly and display it on a smartphone.

Technical objectives:

- design a pre-amplification stage based on the AD620;
- study and implement a Notch filter for 50 Hz power-line interference;
- acquire the conditioned signal using Arduino;
- establish Bluetooth communication with a smartphone;
- develop a MIT App Inventor interface for reception and visualization;
- prepare an architecture that can later support continuous curve recording.

3. Overall Architecture

Simplified functional chain of the prototype.

4. Analog Signal Conditioning

The ECG signal has a very small amplitude and requires conditioning before digital acquisition. The studied circuit includes an AD620 instrumentation amplifier, a right-leg/reference electrode circuit and a Notch filter.

The AD620 was selected because it is suitable for biomedical acquisition systems, provides good common-mode rejection and allows adjustable gain. A simulation included in the original report uses a 20 mV input and obtains

approximately 1800 mV at the output for the considered configuration.

The right-leg circuit is described in the original report as a means of reducing common-mode voltage and contributing to patient protection.

5. 50 Hz Notch Filter

50 Hz power-line interference is a major source of disturbance during ECG acquisition. A Notch filter is therefore used to strongly attenuate this component while preserving the useful frequency content of the signal.

The implemented filter is based on operational amplifiers and its frequency response was studied. Practical tests were carried out using a **TG2000 function generator** and an oscilloscope.

This filtering stage is one of the central technical aspects of the project because it improves the signal before acquisition and transmission.

6. Acquisition, Bluetooth and Mobile Interface

Arduino is used to acquire the conditioned signal. A Bluetooth link transmits the acquired values to a smartphone.

The mobile interface was developed with MIT App Inventor. It receives and displays the transmitted signal. The smartphone was not intended to be only a screen: the initial concept also included using its storage to keep the ECG curves throughout the Holter monitoring period.

The original report refers to both HC-05 and HC-06 in different sections. This revised version therefore uses the generic term **Bluetooth module** to avoid introducing a documentation inconsistency.

7. PCB Fabrication

The PCB fabrication process included preparing the transparent mask, UV exposure of the photosensitive board, development, ferric-chloride etching and component soldering.

The PCB design in Proteus was completed collaboratively within the group. The other technical work described in this report corresponds to the project work carried out during the study.

8. Testing and Results

Practical tests were not performed on a human subject because suitable ECG electrodes and the required testing conditions were not available.

The system was therefore evaluated using **three controlled electrical inputs corresponding to the three electrode inputs**. A voltage/function generator was used to provide controlled signals to the acquisition and conditioning stages.

The tests allowed verification of the acquisition chain, signal conditioning and transmission. The transmitted signal could be displayed on the smartphone through Bluetooth.

These results demonstrate the electronic prototype under the available experimental conditions, but they do not constitute physiological or clinical validation of an ECG system.

9. Limitations and Future Work

Limitations: no human ECG acquisition; controlled electrical signals used for testing; continuous storage of curves on the smartphone was not completed; the prototype was not intended for clinical use.

Future work: perform acquisition using appropriate ECG electrodes and safe testing conditions; improve the analog front-end for real physiological signals; implement continuous recording and smartphone storage; further miniaturize the system for wearable use.

Important: controlled electrical tests verify the electronic chain, but should not be presented as physiological ECG acquisition.

10. General Conclusion

This project implemented the main electronic building blocks of a portable ECG Holter prototype: analog conditioning, amplification, power-line interference reduction, Arduino acquisition and Bluetooth transmission to a smartphone.

The main technical interest of the work lies in the study of the 50 Hz Notch filter and the integration of wireless communication with a mobile interface. The smartphone was considered part of the system from the beginning, first for signal visualization and later, as a planned extension, for storing ECG curves throughout the monitoring period.

The prototype represents an initial biomedical-electronics implementation. The electrical tests verified the functional chain under controlled conditions, while the absence of human testing clearly defines the next step required to move toward real ECG acquisition.

Technical Documentation

The original 2021–2022 report contains the theoretical background, circuit diagrams, simulations, PCB fabrication steps and results documented at the time. This revised version mainly clarifies the introduction, objectives, experimental validation and conclusion so that they better reflect the actual purpose of the prototype.

For GitHub archiving, it is recommended to keep the original report and this revised version as separate documents.